12A — Deep-Dive Feature Audit: UI, Rendering, Serialization & Master Architecture (Part 1 of 4)

Player Agency Boundary (`Observer + Stat/Item Sandbox` vs Forbidden RTS Commands), Edit-Hub Invariant Leaks, Minimap/HUD Desyncs & Root Cause of the `startup_ui` Timeout

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Primary Audited Modules**: `js/ui.js`, `js/main.js`, `js/sect.js` (`SECT.EDITOR`), `index.html`, `test/startup_ui.js`
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini**)
- **Master Roadmap for Final Category 12 (`12A–12D`)**:
  - **`12A` (This Document)**: **Player Agency Philosophy Boundary (`UI.doAction` Audit), Edit-Hub Safety Invariants, UI Truthfulness, Minimap/Loop Optimization & `startup_ui.js` Timeout Fix**
  - `12B`: Canvas 2D Rendering (`render.js`), 43-Path Visuals, Multi-Tier VFX (`Common/T3/T6/T7/Legendary`), `Zero-RNG / Zero-Mutation` Render Purity & Web Audio (`SFX`)
  - `12C`: Save/Load Serialization & Schema Hydration (`SIM.serialize/deserialize`), `40`-Script Order, `build_single.js` Standalone Bundling & Test-Suite Blind Spots
  - `12D`: **Grand Master Index & Multi-Wave Autonomous Patch Execution Protocol** Across All `68` Audit Documents (`01A`–`12D`) for **GPT** & **Gemini**

---

1. `[UI-AGENCY-01]` Philosophical Contamination in Normal Inspection Sheets (`js/ui.js`): Separating Pure Observer Sheets From Sandbox God-Editor Overrides!

1.1 The Core Game Design Law of `JianghuSect 2`

`JianghuSect 2` is a **100% Emergent Living-World Simulation**:

- **What NPCs Own (`100%` Autonomous)**: Choosing cultivation postures, opening meridians, braving tribulations, enacting Sect Charters, convening the 6-Officer Council,

unlocking offices, judging prisoners, deploying formations/barriers, establishing outposts, launching sieges, hunting bounties, and interacting with wandering legends.

- **What the Player Owns (`Observer + God-Sandbox Editor ONLY`)**:
    1. Observing thoughts (`why`), stat breakdowns (`ST.explain`), lineages, Storyline Threads, and Periodic Digests; controlling camera, filters, audio, and simulation speed (`0×..48×`).
    2. Modifying stats/states or spawning items **explicitly inside the Sandbox Edit Hub (`UI.rEdit` / `Menu > Tools`)**—never being forced to click RTS management buttons on normal gameplay drawers just to make systems work!

1.2 Audit of RTS-Style Direct Command Buttons Currently Exposed Outside `UI.rEdit`

Across patches `v3.1.0–v5.8.0`, over **20 features were wired to manual `[data-do]` buttons on `UI.rNpc`, `UI.rSect`, and `UI.rWorldMap` while lacking autonomous NPC triggers**:

- **On `UI.rNpc`**: `set-mantra-stance`, `set-defensive-posture`, `set-support-posture`, `set-forbidden-seal`, `set-focus-constellation`, `open-next-meridian`, `start-tribulation-ritual`, `dual-cultivate`, `vendetta-reclaim`.
- **On `UI.rSect`**: `unlock-specialized-offices`, `enact-sect-charter`, `convene-strategic-council`, `resolve-council-agenda`, `establish-outpost`, `dispatch-supply-convoy`, `set-sect-formation`, `set-barrier-element`, `recharge-barrier`, `set-forbidden-policy`, `invoke-founder-phantom`, `trigger-emergency-support`, `handle-dungeon-prisoner`.
- **On `UI.rWorld` / `UI.rWorldMap`**: `launch-sect-siege`, `collect-vassal-tribute`, `harvest-resource-nodes`, `trigger-alliance-council`, `trigger-secret-realm`, `explore-micro-location`, `claim-bounty`, `run-espionage`.

1.3 Regression-Safe Architectural Migration (`Keeping 100% of Existing Tests Green!`)

1. **Keep `UI.doAction(...)` Handlers Intact for Test & Sandbox Compatibility**: Existing suites (`v320`, `v340`, `v400`, `v530–v580`) invoke `UI.doAction('start-tribulation-ritual', ...)` or `UI.doAction('enact-sect-charter', ...)` programmatically. Keeping `UI.doAction` valid preserves `100%` test compatibility!
2. **Move Manual State-Override Controls Into `UI.rEdit` (`Sandbox Edit Hub`)**:
    - On **`UI.rNpc`, `UI.rSect`, and `UI.rWorldMap`**, replace manual RTS command buttons with **Rich Read-Only Autonomous Status Cards** displaying: *(a)*

Current active posture/charter/policy, (b) Which NPC/Officer holds jurisdiction, and (c) Their live utility reasoning (`why`)—retaining pure camera/filter actions (`focus-tribulation`, `focus-peak-camera`, `select-world-node`, `atlas-layer`, `news-*`, `digest-*`).
- Inside **`UI.rEdit`**, expose clean state-override controls (`clearly badged ✨ Sandbox Override`) for players who *want* to tweak states or test scenarios!

---

2. `[UI-EDIT-02]` Four State-Corruption & Accounting Bugs Inside `SECT.EDITOR` & `Menu > Tools` (`js/sect.js`, `js/ui.js`)!

1. **Editing `cult.path` / `cult.stage` Without Clamping `CULT.maxStage(n)` (`UI.rEdit` `char` Tab)**:
   - If an NPC is at Stage `9` (`XIAN`) and the player edits their path in `UI.rEdit` to `GOD` (`maxStage = 4`), `BODY` (`maxStage = 5`), or any `8`-stage Grand Dao path without clamping `c.stage = Math.min(c.stage, CULT.maxStage(n))`, `CULT.stageInfo(n)` indexes past the stage array or corrupts `normalizedStage`!
   - **Fix**: Whenever `SECT.EDITOR` modifies `cult.path`, `cult.sub`, or `cult.stage`, atomically clamp `c.stage` to `[1, CULT.maxStage(n)]`, re-initialize/sync `c.track`, strip incompatible `c.mantra` / `pathGauge`, refresh `npc.lifespanMax`, call `HC.bump(n)`, and call `MARTIAL.reconcilePassives(n)`!
2. **`Menu > Tools` (`data-do="tool"`, `stones`/`grain`/`fame`) Must Record Player-Intervention Provenance (`v2.6.0` Bucket #4)**:
   - Adding `+1,000` stones via `[data-do="tool"][data-arg="stones"]` (`verified in startup_ui.js:68`) must tag the ledger entry with `{ playerIntervention: true, spawned: true }` so `EXPED.computePeriodDigest` records it cleanly in **Account #4 (การแทรกแซงของผู้เล่น)** rather than polluting natural sect revenue or leaving an unexplained closing-balance gap!
3. **Exclude `W._editUndoStack` From `SIM.serialize(W)` (`localStorage` Quota Safety)**:
   - Storing full-NPC JSON undo snapshots inside `W._editUndoStack` must never be serialized into `localStorage.setItem('jhs_auto', ...)` (`cap W._editUndoStackat8 entries in RAM only`).
4. **Character Sheet `ST.explain` vs `ST.get` Truthfulness (`09A` Cross-Link)**:
   - `UI.rNpc` renders `STATS.explain(n)` while combat uses `STATS.get(n)`—unifying their modifier helper guarantees the player never sees

fake `+8% ATK` on `MECH_PUPPET 'OVERDRIVE'` while the NPC actually gets `+0% ATK`!

---

3. `[UI-LOOP-03]` Root Cause & Exact Fix for the `test/startup_ui.js` Overlay Timeout, `39`-Path Minimap Color Blindness & Paused Redraw Drain!

3.1 Solving the Recorded `startup_ui.js` Timeout (`test/startup_ui.js:97–103` & `index.html` / `js/ui.js`)!

In the `JS2-v2.6.0` regression log (`161/162 PASS, 1 startup_ui overlay TIMEOUT`), inspect test #7 of `startup_ui.js` (`lines 97–102`):

```javascript
await test('a successful import dismisses an open startup overlay', async page =>
{
  const payload = await savePayload(page, 'startup-overlay-import');
  await page.setInputFiles('#impFile', { name: 'saved.json', mimeType:
'application/json', buffer: Buffer.from(payload) });
  await page.waitForFunction(() => G.W && G.W.seed === 'startup-overlay-import');
  assert.strictEqual(await page.locator('#start').isVisible(), false, 'A loaded
world must not remain covered by startup');
});
```

- **Exact Root Cause**:
  1. On initial page boot (`UI.showStart()`), `#start` is visible and `#sheet` (`UI.rMenu`) has **not** been opened yet! If `<input type="file" id="impFile">` is only injected dynamically when `UI.rMenu` renders (rather than existing statically in `index.html` / `UI.init`), `page.setInputFiles('#impFile', ...)` hangs for `30,000 ms` waiting for `#impFile` to exist!
  2. Furthermore, when `#impFile`'s `change` listener calls `G.load(text)`, it must explicitly dismiss **both** `#sheet` (`UI.close()`) **and** `#start` (`UI.hideStart()` / `$('start').style.display = 'none'`)!
- **Exact Fix**: Mount a single persistent hidden `<input id="impFile" type="file" accept=".json,application/json" style="display:none">` once in `index.html` / `UI.init(G)` wired to `G.load(txt)` + `UI.hideStart()` + `UI.close()`, turning `161/162` into **`162/162 PASS` (`0` timeouts!)**!

3.2 `39` Cultivation Paths Hardcoded to Generic Cyan on Minimap (`js/main.js:84`)

- `main.js:84` checks only `BODY`, `GOD`, `DEMON`, and `HEART_SOUL`, painting all other **39 Major Paths** as `#4fb2f2`!
- **Fix**: Resolve `(Render.PATH_VISUAL_SIGNATURES && Render.PATH_VISUAL_SIGNATURES[n.cult.path]?.primary) || '#4fb2f2'` in `O(1)`!

3.3 Unnecessary `8.3 FPS` Full Canvas Redraw When Paused (`G.speed === 0`, `js/main.js:69`)

- When `G.speed === 0`, only redraw `R.draw(ts)` when `R._camDirty || R._uiDirty` (set on pan, zoom, resize, selection, or Edit-Hub change) instead of burning battery every `120 ms` on an unchanging frame!